

HDOC Day-12

1. Problem Statement

Write a **menu-driven C program using switch-case** to input a positive integer and perform the following operations:

1. Find the **sum of the first and last digits**.
2. Find the **product of the first and last digits**.

Example:

Number = 12345

First digit = 1

Last digit = 5

Sum = $1 + 5 = 6$

Product = $1 \times 5 = 5$

2. Algorithm

1. Start.
2. Input a positive integer num.
3. Display the menu:
 - 1. Sum of first and last digit
 - 2. Product of first and last digit
4. Input the user's choice.
5. Find the last digit using:
 $\text{last} = \text{num} \% 10$
6. Store num in a temporary variable temp.
7. Divide temp by 10 repeatedly until it becomes less than 10.
8. The remaining value of temp is the first digit.

9. Use switch(choice):

- **Case 1:** Calculate sum = first + last and display it.
- **Case 2:** Calculate product = first * last and display it.
- **Default:** Display "Invalid choice".

10. Stop.

3. Test Case Table

Test Case Number	Choice	Expected Output	Actual Output	Result
------------------	--------	-----------------	---------------	--------

1	12345	1	Sum = 6	Sum = 6	Pass
2	12345	2	Product = 5	Product = 5	Pass
3	5678	1	Sum = 13	Sum = 13	Pass
4	5678	2	Product = 40	Product = 40	Pass
5	909	1	Sum = 18	Sum = 18	Pass
6	909	2	Product = 81	Product = 81	Pass

4. C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int num, temp, first, last, choice;
```

```
    int sum, product;
```

```
    printf("Enter a positive integer: ");
```

```
scanf("%d", &num);
```

```
if (num <= 0)
```

```
{
```

```
    printf("Please enter a positive integer.\n");
```

```
    return 0;
```

```
}
```

```
printf("\n1. Find sum of first and last digit\n");
```

```
printf("2. Find product of first and last digit\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
last = num % 10;
```

```
temp = num;
```

```
while (temp >= 10)
```

```
{
```

```
    temp = temp / 10;
```

```
}
```

```
first = temp;
```

```
switch (choice)
```

```
{
```

case 1:

```
sum = first + last;  
printf("First digit = %d\n", first);  
printf("Last digit = %d\n", last);  
printf("Sum = %d\n", sum);  
break;
```

case 2:

```
product = first * last;  
printf("First digit = %d\n", first);  
printf("Last digit = %d\n", last);  
printf("Product = %d\n", product);  
break;
```

default:

```
printf("Invalid choice\n");
```

```
}
```

```
return 0;
```

```
}
```

```
kali@kali:~/Desktop$ gcc hdoc.c -o hdoc
kali@kali:~/Desktop$ ./hdoc
Enter a positive integer: 12345

Menu:
1. Find Sum of first and last digits
2. Find Product of first and last digits
Enter your choice: 1
Sum = 6

kali@kali:~/Desktop$ ./hdoc
Enter a positive integer: 12345

Menu:
1. Find Sum of first and last digits
2. Find Product of first and last digits
Enter your choice: 2
Product = 5

kali@kali:~/Desktop$ ./hdoc
Enter a positive integer: 9876

Menu:
1. Find Sum of first and last digits
2. Find Product of first and last digits
Enter your choice: 1
Sum = 15

kali@kali:~/Desktop$ █
```

Result: The program was compiled and tested against the prepared test cases. All test cases produced the expected results.